

Supplementary Material: Benchmarking LLM Integration Strategies for MCDA-Assisted Household Energy Decision Support

Ahaan Nigam and River Huang

S1. Extended Methods

The four models and their exact OpenRouter identifiers are GPT-OSS-20B (`openai/gpt-oss-20b:exacto`, reasoning, low effort), Qwen 3.5 9B (`qwen/qwen3.5-9b:exacto`, non-reasoning), DeepSeek V4 Flash (`deepseek/deepseek-v4-flash:exacto`, hybrid reasoning), and Gemini 3.5 Flash (`google/gemini-3.5-flash:exacto`, reasoning, minimal effort). The `:exacto` suffix is an OpenRouter provider-routing variant that restricts serving to providers meeting a fixed accuracy standard; it is part of the model identifier and is required to reproduce these runs.

At least two alternatives share a weighted score in 3.2% of Gemini’s $\mathcal{A}_D$ scenarios, 5.9% of DeepSeek’s, 12.6% of GPT-OSS’s and 21.3% of Qwen’s, and in 2.7%, 19.1%, 6.4% and 13.9% of the same four models’ $\mathcal{A}_E$ scenarios. All three alternatives carrying identical scores on all four criteria is rarer, between 0.0% and 1.2% in every model–architecture cell, so the tie-break rule usually separates alternatives that already differ on at least one criterion.

0.1 Full Prompt Texts

Full verbatim text of the system and extraction prompts, in the order the architectures appear in the framework.

0.2 $\mathcal{A}_D$ System Prompt

System prompt:

You are an expert household decision analyst specializing in Multi-Criteria Decision Analysis (MCDA).

You consistently utilize all information given in the scenario context. Score alternatives on four criteria using the

HVAC:

- energy_cost: good when the setpoint demands little from the system given outdoor conditions; moderate when the system demands more; poor when the system demands a lot.
- environmental: good when the system runs efficiently; moderate when runtime and load are average; poor when high.
- comfort: good when the setpoint feels comfortable for the outdoor temperature; moderate when slightly outside typical range; poor when too far from comfort.
- practicality: good when the setpoint is easy for the user and system to maintain without strain; moderate when somewhat difficult; poor when very difficult.

Appliance (scheduling):

- energy_cost: good during off-peak rate hours; moderate in shoulder periods; poor when scheduled during peak pricing.
- environmental: good when grid emissions are low (typically overnight); moderate during shoulder hours; poor when high.
- comfort: good when the appliance is available with little or no delay; moderate when availability is pushed later than desired; poor when too late.
- practicality: good when scheduling is easy to remember and socially unobtrusive; moderate when timing requires some effort; poor when very inconvenient.

Shower (duration):

- energy_cost: good when short; worsens continuously as duration increases.
- environmental: good when water volume used is low; worsens continuously as duration increases.
- comfort: good near a typical shower length; poor when too short to feel adequate or long enough to feel wasteful.
- practicality: good when the duration is sustainable as a daily habit; poor when too short to realistically maintain.

Return ONLY: {"energy_cost": X, "environmental": X, "comfort": X, "practicality": X} where each X is between 0.0 and 1.0.
The user prompt includes the other alternatives as reference so that it doesn't assign the same scores to all alternatives.

0.3 $\mathcal{A}_E$ System Prompt

System prompt:

You are an expert household decision analyst specializing in Multi-Criteria Decision Analysis (MCDA).

You consistently utilize all information given in the scenario context. Score alternatives on four criteria using the

1. Energy Cost: Lower energy costs = higher score.
2. Environmental Impact: Lower emissions = higher score.
3. Comfort: Higher user comfort = higher score.
4. Practicality: Easier to implement/maintain = higher score.

Return ONLY: {"energy_cost": X, "environmental": X, "comfort": X, "practicality": X} where each X is between 0.0 and 1.0.

0.4 $\mathcal{A}_H$ Extraction Prompt

Extraction prompt:

You are a household-systems engineer. You base all assumptions off of given information. Understand how each parameter

SCENARIO:

{scenario_text}

QUESTION: {question}

INSTRUCTIONS:

1. Identify the Decision Type: HVAC, Appliance, or Shower.
2. Estimate ONLY the parameters listed for that type below.
3. Every listed parameter is mandatory. If a value is not stated, it is mandatory to reasonably estimate it from the scenario.
4. Return ONLY valid JSON in the exact structure shown.

For HVAC decisions:

```
{"decision_type": "HVAC",
  "calculator": "HVACGroundTruthCalculator",
  "parameters": {"r_value": <number>,
    "seer": <number>,
    "hvac_age": <number>,
    "occupancy_context": "occupied_all_day |
      unoccupied.<hours> | occupied_sleep"}}
```

For Appliance decisions:

```
{"decision_type": "Appliance",
  "calculator": "ApplianceGroundTruthCalculator",
  "parameters": {"appliance": "Dishwasher | Washer | Dryer",
    "kwh_per_cycle": <number>,
    "baseline_time": "<the time it currently is, e.g. 7pm, 8am>"}}
```

For Shower decisions:

```
{"decision_type": "Shower",
  "calculator": "ShowerGroundTruthCalculator",
  "parameters": {"gpm": <number>,
    "tank_size": <number>,
    "water_heater_temp": <number>}}
```

Return ONLY the JSON, no explanation.

0.5 Retrieved Exemplar Rendering

A retrieved HVAC exemplar (engineering parameters and ground-truth scores included) is rendered as:

RELEVANT SIMILAR SCENARIOS WITH EXPERT SCORES:

Example 1:

- Question: I'm heading out for a quick errand, what AC temperature should I set for the townhouse?
- Location: King of Prussia, PA
 - Outdoor Temp: 93 deg F
 - Square Footage: 2200 sqft
 - Insulation: Medium
 - Household Size: 5 occupants

- Housing Type: Townhouse
- House Age: 11-15 years
- R-Value: 17
- SEER: 15
- HVAC Age: 12 years
- Utility Budget: \$285/month

Expert scores:

- * 75 F: Energy Cost: 0.5/1, Environmental: 0.5/1,
Comfort: 0.9/1, Practicality: 0.8/1
| MAVT: 0.61, Rank: 2
- * 78 F: Energy Cost: 0.5/1, Environmental: 0.5/1,
Comfort: 0.8/1, Practicality: 0.8/1
| MAVT: 0.62, Rank: 1
- * 81 F: Energy Cost: 0.5/1, Environmental: 0.5/1,
Comfort: 0.5/1, Practicality: 0.8/1
| MAVT: 0.58, Rank: 3

Use these examples as reference, but score based on the specific scenario above.

0.6 RAG Corpus Description

The retrieval-augmented generation (RAG) corpus consists of 90 unique scenarios (35 HVAC, 35 Appliance, 20 Shower) that are disjoint from the 195 test scenarios. Each RAG scenario has three alternatives, yielding 270 total rows across the three source files (HVACRagScenarios.xlsx, ApplianceRAGScenarios.xlsx, ShowerRAGScenarios.xlsx).

Scenarios were generated from the same parameter distribution as the test set but kept separate to prevent any test scenario from appearing in the retrieval index. A combinatorial audit ensured that no parameter combination was underrepresented: each categorical cross-cell (insulation tier $\times$ housing type, appliance type $\times$ flow-rate band, etc.) contained at least one scenario, and the 90 RAG scenarios were drawn from this audited matrix. The 195 test scenarios were built and rebalanced against the same audited matrix, so that every categorical combination appears at least once in each decision type’s test set.

The retrieval index was built using the `sentence-transformers/all-MiniLM-L6-v2` embedding model (Reimers and Gurevych, 2019), producing 384-dimensional dense vectors. At query time, the single most similar scenario ($k = 1$) is retrieved by Euclidean (L2) distance, Chroma’s default metric, and rendered as a full worked example in the LLM prompt. The embedding string includes all household-reported parameters (location, square footage, insulation tier, household size, housing type, utility budget, outdoor temperature, house age) plus band labels for age and flow rate, but excludes engineering parameters and ground-truth scores, which appear only in the display metadata of the retrieved exemplars.

The disjointness of test and RAG sets was verified by construction (the rebuild script exports them from separate master-sheet partitions).

0.7 Parameter Ranges and Band Labels

Each insulation tier spans a range of R-values drawn from standard U.S. residential wall assemblies (Commission, 2013; U.S. Environmental Protection Agency and U.S. Department of Energy, 2023): Poor scenarios use R-8 to R-12 (minimal 2 $\times$ 4 cavity fill, below current code), Medium scenarios use R-13 to R-18 (standard 2 $\times$ 4 batt through 2 $\times$ 6 mixed), and Good scenarios use R-19 to R-24 (2 $\times$ 6 framing and above). Consecutive tier midpoints differ by 55% (R-10 to R-15.5) and 39%

(R-15.5 to R-21.5) in wall R-value. SEER ratings in the test corpus span 8–22, covering pre-1992 units (SEER 8–10), standard-efficiency replacements (SEER 13–14), and high-efficiency models (SEER 16–22), consistent with the ENERGY STAR Most Efficient threshold (U.S. Environmental Protection Agency, 2022).

The appliance age presented to the LLM architectures is a band label (1–3, 4–6, 7–9, 10–12, 13–17, and 28–32 years in the test corpus), derived from the scenario’s raw integer age by a single shared helper so that the label is byte-identical on the query and index sides of retrieval. The scenario sheets retain both the raw age and an independent per-cycle energy value (kWh/cycle) drawn from the ENERGY STAR certified-products database, and the ground-truth calculator scores kWh/cycle directly without reference to age. Within a single appliance type the band and the per-cycle energy are only moderately associated (Spearman’s $\rho = 0.36$ for dishwashers, 0.38 for dryers, and 0.69 for washing machines), so two appliances sharing a band can differ materially in energy draw—a 7–9 year dryer ranges from 2.40 to 3.50 kWh/cycle across the corpus.

For the Flow Rate parameter, the scenario’s numeric GPM is bucketed into three qualitative labels: `low_flow` (≤ 2.0 GPM, including EPA WaterSense certified showerheads (Agency, 2018)), `standard` (2.5–3.0 GPM, at or near the Energy Policy Act of 1992 federal maximum of 2.5 GPM (Agency, 2018)), and `high_flow` (> 3.0 GPM). The label is present in the LLM test scenarios only, and is mapped from those ranges to the actual numeric GPM from the actual scenario.

0.8 Worked Example: $\mathcal{A}_H$ HVAC Scenario

This subsection traces one HVAC scenario end-to-end through the $\mathcal{A}_H$ pipeline. The full step-by-step arithmetic (thermal load, EER conversion, energy/cost/emissions, budget penalty, comfort and practicality scores, value-function transformation) is reproducible by running `Ground Truth Calculators/HVACGroundTruthCalculator.py` in the project repository on the scenario below; only the inputs and the final result are shown here.

The scenario: Philadelphia, PA; outdoor temperature 95 °F (cooling season); 2,200 sq ft single-family home, medium insulation, house age 15 years; household size 4; \$200/month utility budget; alternatives of 72 °F, 76 °F, and 80 °F. $\mathcal{A}_H$ ’s extraction step returns $R = 15$, SEER = 13, hvac.age = 10, and `occupancy_context = occupied_all_day`, each validated against the physically admissible bounds before being merged with the household-reported fields above and the fixed electricity rate of \$0.19/kWh to produce the full scenario dict passed to `HVACGroundTruthCalculator`.

Carrying that scenario through the HVAC calculator equations above for all three alternatives yields:

Setpoint	v_{cost}^*	v_{env}	v_{comfort}	v_{prac}	MAVT s_j
76 °F	0.11	0.30	1.00	0.85	0.466
80 °F	0.17	0.37	0.67	0.85	0.441
72 °F	0.06	0.24	0.67	0.78	0.352

0.9 GPT-OSS Extraction Failure Analysis

The $\mathcal{A}_H$ architecture validates each extracted JSON before passing it to the ground-truth calculator. A scenario is recorded as an extraction failure when: the LLM response is not valid JSON; a required parameter is absent; a numeric parameter parses as non-finite or falls outside its physically admissible range; or the extracted decision type does not match the scenario’s known type.

GPT-OSS 20B exhibited a 33.4% extraction failure rate on $\mathcal{A}_H$ HVAC scenarios (12.0% across all decision types), triggered entirely by invalid parameters. All of its $\mathcal{A}_H$ failures were HVAC

scenarios; Appliance and Shower extraction succeeded in every run. In a typical failure the model returns well-formed JSON but estimates parameters outside the validation bounds—for example, a SEER rating of 5 against a lower bound of 6.0. The validation layer detects the out-of-range value, flags the scenario as an extraction failure, and assigns the sentinel value (1928) to all four criterion scores for that scenario.

Table S1.1 breaks the failure counts down by failure mode. The most common failure mode for $\mathcal{A}_D$ and $\mathcal{A}_E$ is `EXTRACTION_INVALID_JSON` (model returns unparseable output), while for $\mathcal{A}_H$, all failures concentrate in `FAILED_EXTRACTION_INVALID_PARAMETERS` (extracted numeric parameter falls outside the physical validation bounds), driven almost entirely by GPT-OSS on HVAC scenarios.

Table S1.1: Nonzero-firing failure mode counts per architecture (5-run total across all models).¹

Mode	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$
<code>EXTRACTION_INVALID_JSON</code>	5	11	0
<code>FAILED_MISSING_SCORE</code>	1	0	0
<code>FAILED_EXTRACTION_INVALID_PARAMETERS</code>	0	0	120

0.10 Pennsylvania Time-of-Use Rate Structure

Table S1.2 lists the six Pennsylvania utilities used by the appliance ground-truth calculator, with their TOU peak/off-peak windows and energy rates (\$/kWh). All scenarios use weekday summer rates year-round; seasonal peak-window shifts (e.g., PPL winter 4–8pm vs. summer 2–6pm) and weekend off-peak pricing are not modeled.

PECO (Rate R-TOU) (Company, 2021) and PPL (Rate RTS) (PPL Electric Utilities, 2025) use a 4-hour afternoon peak window (2–6pm weekdays). The three FirstEnergy utilities (West Penn, Penelec, MetEd) use a 7-hour peak window (2–9pm weekdays), with rates derived from the Jun-2025 Price-to-Compare (PTC) multiplied by approved multipliers (FirstEnergy Pennsylvania Electric Company, 2025, 2026). Duquesne uses a flat default rate (Rate RS) (Duquesne Light Company, 2026). Shifting a dishwasher cycle from peak to off-peak cuts its per-cycle cost by 49–76% across these five time-of-use utilities.

0.11 Realized Criterion Swing

Table S1.3 reports, for each criterion and decision type, the mean over scenarios of the nominal weight times the within-scenario score range—the quantity each criterion actually contributes to the spread in MAVT score within a choice set, as distinct from its nominal weight.

0.12 Full Sensitivity Analysis Results

Table S1.4 gives Kendall’s τ for every model, architecture and weight vector. Table S1.5 gives Top-1 accuracy for the same cells. The weight vectors are the design vector (0.30, 0.35, 0.20, 0.15), the equal-weight vector (0.25, 0.25, 0.25, 0.25), and the entropy- and MEREC-derived vectors of Appendix B, each applied pooled across the corpus and per decision type. No value in either table is pooled across models.

Table S1.2: Pennsylvania utility TOU rate structure. Peak window is 24-hour clock (start, end). All rates in \$/kWh. The residential electricity rate of \$0.19/kWh used for HVAC cost normalization is the statewide bundled baseline (Administration, 2025; Pennsylvania Public Utility Commission, 2025).

Utility	Peak Window	Peak Rate	Off-Peak Rate
PECO	2–6pm	0.3200	0.0760
PPL	2–6pm	0.1600	0.0700
West Penn	2–9pm	0.1720	0.0880
Penelec	2–9pm	0.1850	0.0930
MetEd	2–9pm	0.2030	0.1000
Duquesne	2–9pm	0.1375	0.1375

Criterion	Nominal Weight	HVAC	Appliance	Shower
Energy Cost	0.30	0.027	0.066	0.108
Environmental Impact	0.35	0.032	0.008	0.128
Comfort	0.20	0.086	0.098	0.070
Practicality	0.15	0.015	0.095	0.053

Table S1.3: Mean within-scenario swing (nominal weight $\times$ within-scenario score range) by criterion and decision type.

0.13 Ablation Data Provenance

The RAG and prompt-sensitivity ablations reported in this section were collected against an earlier shower comfort function. That function scored the water-heater setpoint, a storage temperature, against a 110–130 °F no-penalty band taken from delivery-side comfort and scald thresholds. The shipped function uses 120–140 °F, anchored instead to CDC storage guidance: below 120 °F a tank cannot deliver the 105 °F target with mixing margin and sits inside the *Legionella* amplification range, and above 140 °F adds standby loss without comfort benefit. The main manuscript’s Calculator Simplifications subsection discusses the correction.

We recomputed the shower reference under the corrected band before deciding whether to re-collect. The setpoint enters comfort as a per-scenario constant shared by all three alternatives, so it moves a ranking only through the criterion balance. Across the 60 test shower scenarios the correction changes the reference winner in none and the full reference ranking in none. In the 20-scenario shower RAG corpus it changes 1 winner and 2 rankings. Both ablation suites test whether a configuration preserves the architecture ordering, and the narrowest ordering margin either suite resolves is 0.13 in Kendall’s τ , so a correction that moves no test reference winner cannot reverse either suite’s conclusions. Additionally, 799 of 14,400 $\mathcal{A}_H$ Shower comfort cells (5.5%) failed to reproduce under the uncorrected calculator because shipped scores omitted cold-weather duration elasticity. Realignment elasticity and applying the 120–140 °F band yields $\tau = 0.8117$ versus shipped 0.8111, confirming the gap is rank-neutral and affects score errors only. The headline results in the main text use the corrected reference throughout.

Sensitivity analysis: Kendall’s τ by model under reweighting

Weight vector	Gemini			DeepSeek			GPT-OSS			Qwen		
	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$
Design (baseline)	0.176	0.310	0.923	0.144	0.307	0.897	0.041	0.272	0.897	0.010	0.208	0.880
Equal	0.312	0.389	0.911	0.115	0.311	0.909	0.081	0.267	0.877	−0.028	0.225	0.879
Entropy pooled	0.197	0.331	0.919	0.126	0.314	0.873	0.039	0.279	0.880	0.004	0.197	0.848
Entropy per-type	0.170	0.336	0.917	0.088	0.347	0.896	0.043	0.280	0.885	−0.007	0.218	0.886
MEREC pooled	0.655	0.632	0.956	0.220	0.410	0.963	0.233	0.329	0.953	0.000	0.267	0.963
MEREC per-type	0.602	0.625	0.929	0.308	0.367	0.930	0.163	0.372	0.900	0.041	0.248	0.921
MEREC HVAC vector	0.676	0.600	0.985	0.333	0.413	0.984	0.275	0.427	0.976	0.064	0.282	0.984

Both the reference ranking and the architecture ranking are recomputed under each vector, because a MAVT reference is defined by its weights. Bold marks the two cells in which $\mathcal{A}_D$ exceeds $\mathcal{A}_E$; both occur for Gemini under MEREC vectors. Top-1 accuracy for the same cells appears in Table S1.5 below.

Table S1.4: Kendall’s τ by model and architecture under seven weight vectors. No value is pooled across models.

Table S1.5: Sensitivity analysis: Top-1 accuracy (%) by model and architecture under seven weight vectors.

Weight vector	Gemini			DeepSeek			GPT-OSS			Qwen		
	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$
Design (baseline)	36.2	48.7	93.1	36.7	53.5	90.8	33.3	47.4	91.7	30.0	46.6	89.7
Equal	50.0	55.0	94.4	36.8	52.5	93.7	37.7	48.5	92.7	28.8	46.2	89.5
Entropy pooled	37.5	48.9	93.7	35.5	54.2	89.8	33.7	48.5	91.7	29.6	46.5	88.4
Entropy per-type	39.0	52.1	94.6	34.2	56.0	92.4	34.7	49.3	92.4	30.1	46.1	92.7
MEREC pooled	75.7	74.4	95.3	43.6	57.4	95.5	46.9	53.9	94.1	32.3	49.8	97.5
MEREC per-type	69.1	76.1	93.8	49.4	52.6	94.5	41.5	57.0	92.3	34.5	50.0	92.2
MEREC HVAC vector	80.4	71.5	98.5	53.6	56.5	97.9	49.1	61.1	96.9	37.1	51.6	99.2

Percentage of the 195 test scenarios whose top-ranked alternative matches the reference. Bold marks cells in which $\mathcal{A}_D$ exceeds $\mathcal{A}_E$; these are the same two Gemini MEREC cells that reverse on Kendall’s τ .

0.14 RAG Ablation: Full Per-Configuration Table

0.15 RAG Ablation: Test-Family Bookkeeping

A Friedman test is run per metric (Kendall’s τ , Top-1 accuracy, MAE, RMSE) across the seven LLM configurations within each model, giving a sixteen-test family of four models by four metrics; the offline nearest-neighbor baseline has no per-model configuration and so enters the descriptive bootstrap reported above rather than any omnibus. Holm correction is applied across this family before any omnibus result is treated as significant, and post-hoc pairwise Wilcoxon signed-rank tests, themselves Holm-corrected within their own pairwise family, are computed only for the model–metric cells whose corrected omnibus remains significant.

0.16 Run-to-Run Variance: Full Distributions

MAE and Top-1 distribution shapes across runs, and the per-model MAE breakdown, are omitted here because Table S1.15 (0.26) already reports mean $\pm$ std for every metric.

Configuration	DeepSeek	Gemini	GPT-OSS	Qwen
<i>Kendall's τ</i>				
Exemplars w/o hidden params	0.391	0.437	0.175	0.273
$k=3$ control setting (all-MiniLM-L6-v2)	0.489	0.387	0.187	0.154
Alt. embedding	0.346	0.517	0.227	0.165
Retrieval $k = 1$	0.182	0.377	0.237	0.306
Retrieval $k = 5$	0.342	0.431	0.112	0.234
Random exemplars ($k = 3$)	0.063	0.177	0.022	0.276
Descriptions w/o scores or ranks	0.073	0.204	0.061	0.071
<i>MAE</i>				
Exemplars w/o hidden params	0.087	0.078	0.106	0.099
$k=3$ control setting (all-MiniLM-L6-v2)	0.088	0.070	0.105	0.102
Alt. embedding	0.088	0.071	0.113	0.104
Retrieval $k = 1$	0.098	0.097	0.118	0.097
Retrieval $k = 5$	0.084	0.069	0.102	0.088
Random exemplars ($k = 3$)	0.127	0.139	0.163	0.141
Descriptions w/o scores or ranks	0.131	0.143	0.158	0.151
<i>Nearest-neighbor (offline, no LLM)</i>				
Kendall's τ / MAE	0.001 (95% CI $[-0.172, 0.175]$) / 0.101			

Evaluated on the full 90-scenario RAG corpus (35 HVAC / 35 Appliance / 20 Shower) under leave-one-out retrieval, with all four models. Values are per-model bootstrap point estimates; the corresponding per-model 95% percentile bootstrap intervals are in `Analysis/RAG_Ablation/rag_ablation_bootstrap_ci.xlsx` in the repository. Earlier versions of this table reported a single cross-model figure per configuration; those pooled estimates are withdrawn, because pooling reversed the random-exemplar comparison for Qwen and hid a significant $k=1$ against $k=3$ difference for DeepSeek. Nearest-neighbor assigns retrieved scenario scores directly without LLM rescoring and uses no LLM, so it has no per-model row. Across all LLM configurations 7,560 scoring calls were made ($4 \text{ models} \times 7 \text{ configurations} \times 90 \text{ scenarios} \times 3 \text{ alternatives}$) with 2 failures (0.03%), both on DeepSeek.

Table S1.6: RAG ablation: Kendall's τ and MAE by configuration.

0.17 Statistical Significance Tests

All tests use per-scenario metrics computed from the 195 test scenarios, with five independent runs per architecture-model combination.

0.18 Pairwise Wilcoxon Tests, Reported Per Model

Wilcoxon signed-rank tests compare each pair of adjacent architectures ($\mathcal{A}_D$ vs. $\mathcal{A}_E$ and $\mathcal{A}_E$ vs. $\mathcal{A}_H$) on each of the three primary metrics (τ , MAE, Top-1), applied within each of the four models. All 24 per-model comparisons remain significant under Holm correction, whether corrected across the 24 tests reported here or across the full 56-test family. The largest adjusted values are DeepSeek's and Gemini's $\mathcal{A}_D$ vs. $\mathcal{A}_E$ tests on τ (both $p_{\text{Holm}} = 0.0062$) and Gemini's on Top-1 (0.0036); the other 21 fall at or below 1.6×10^{-4} , and the smallest is 5.2×10^{-31} . GPT-OSS contributes $n = 194$ rather than 195 on the $\mathcal{A}_E$ vs. $\mathcal{A}_H$ comparison, one scenario dropping from the paired set.

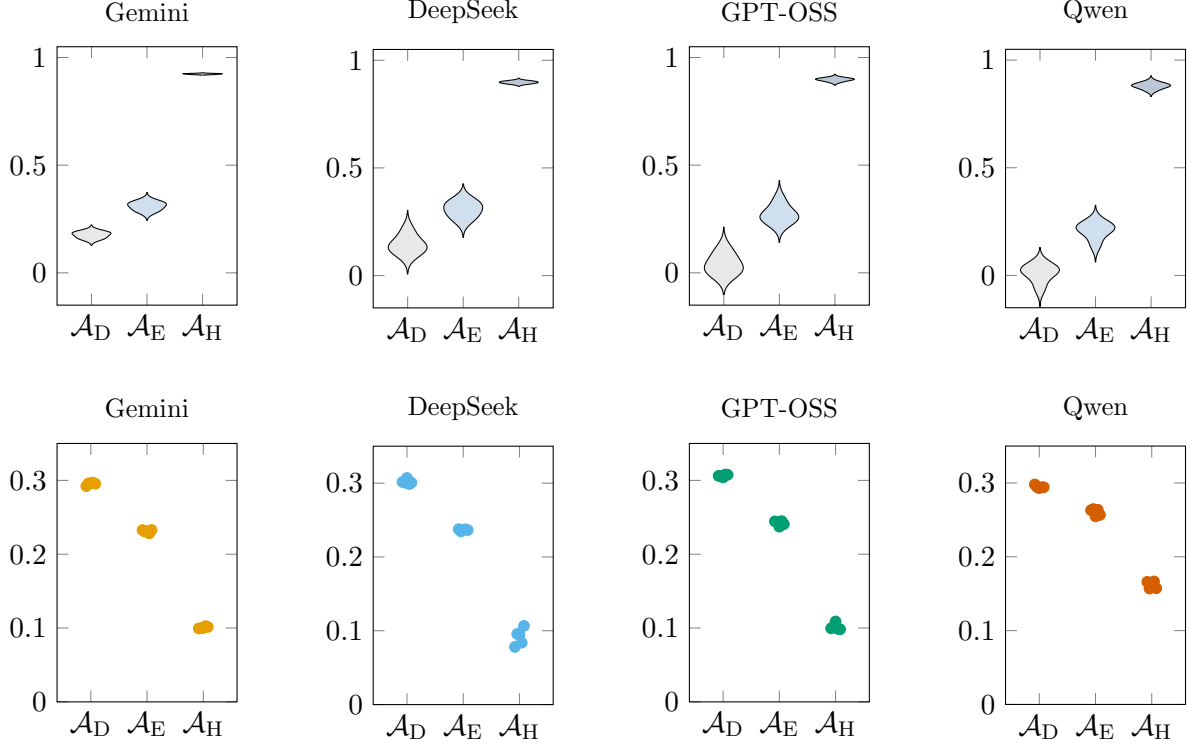


Figure S1.2: Strip plots of Overall RMSE across five runs.

Two comparisons carry heavy tie loads, both of them Gemini’s $\mathcal{A}_D$ vs. $\mathcal{A}_E$: 58 of 195 paired differences are exactly zero on τ , and 102 of 195 on Top-1. Each reports a median paired difference of 0.0000 while the test stays significant, because the signed-rank statistic is computed over the non-tied pairs alone. Read Gemini’s effect size for that comparison from Cliff’s δ in Table S1.7 rather than from the median. Every test uses the asymptotic approximation with tie and continuity corrections; the exact method applies to no cell, since each either exceeds 25 non-zero pairs or carries ties. Per-test Z , median paired difference, tied-pair counts, and both Holm-corrected p -values are in `paper/per_model_pvalues.csv` in the repository.

0.19 Effect Sizes

Table S1.7 reports Cliff’s δ for each pairwise comparison.

0.20 Bootstrap Confidence Intervals

Confidence intervals resample scenarios, not runs. Each of the 10,000 replicates draws 195 scenarios with replacement and carries all five runs of a drawn scenario together, so an interval covers the variation a different draw of households produces rather than the variation a different sampling seed produces. Scenarios are the unit a reader generalizes over, so the resampling treats them as the cluster. Table S1.8 reports paired architecture differences, drawing one scenario index and scoring both arms on it.

Table S1.7: Cliff’s δ effect sizes for pairwise architecture comparisons, shown per model.

Comparison	Model	τ	MAE	Top-1
$\mathcal{A}_D$ vs. $\mathcal{A}_E$	DeepSeek	−0.159 (small)	0.494 (large)	−0.257 (small)
	Gemini	−0.129 (negligible)	0.400 (medium)	−0.157 (small)
	GPT-OSS	−0.273 (small)	0.489 (large)	−0.207 (small)
	Qwen	−0.246 (small)	0.319 (small)	−0.305 (small)
$\mathcal{A}_E$ vs. $\mathcal{A}_H$	DeepSeek	−0.753 (large)	0.884 (large)	−0.596 (large)
	Gemini	−0.592 (large)	0.778 (large)	−0.495 (large)
	GPT-OSS	−0.780 (large)	0.843 (large)	−0.661 (large)
	Qwen	−0.835 (large)	0.749 (large)	−0.721 (large)

0.21 Prompt Ablation: Full Results Table

Table S1.9 reports Kendall’s τ and Top-1 accuracy for every architecture, model, and prompt variant.

0.22 Prompt Ablation Significance

Gemini’s prompt-ablation cells use three runs against the five or ten the other three models use; the significance tests are over $n = 195$ scenarios rather than over runs, and Gemini’s run-to-run standard deviation of 0.008–0.021 is the lowest of the four models, so three runs resolve its cell means at the precision reported. The no_anchors variant runs on ten repetitions instead of five for DeepSeek, GPT-OSS, and Qwen, versus five for cot_scaffold and scale_0.10; Gemini runs three repetitions on every variant.

Table S1.10 reports the Friedman omnibus tests for $\mathcal{A}_E$, and Table S1.11 reports the pairwise comparisons that reach significance under the two-stage Holm correction with a fully reported effect size.

0.23 Parameter-Source Ablation Significance

Table S1.12 reports the parameter-source ablation in full: for each model, ranking accuracy with the scenario’s true hidden parameters (the ceiling), the LLM’s extracted parameters, and a single corpus-median constant per parameter (no per-scenario inference).

Significance for the parameter-source ablation is tested within each model rather than pooled across models. Four models times three metrics (Kendall’s τ , Top-1 accuracy, MAE) gives twelve Holm-corrected Friedman omnibus tests, with post-hoc Wilcoxon testing restricted to the model–metric cells whose corrected result remains significant.

Friedman tests confirm the three conditions (true, LLM-extracted, corpus-median) differ from one another across the 195 scenarios for every model on every metric (χ^2_2 ranging from 52.5 to 305.3, all $p_{\text{Holm}} < 10^{-11}$ across the twelve-test family). The omnibus is a weak gate in this design rather than a finding: true_params reproduces the ground truth by construction and default_params is a fixed constant, so the contrast between them carries no model output and returns the same p-value for all four models. The claim rests on the post-hoc comparison instead. Post-hoc Wilcoxon signed-rank tests show the extracted condition beats the corpus-median condition on Kendall’s τ for all four models ($p_{\text{Holm}} < 10^{-6}$, Cliff’s δ between −0.22 and −0.24, small) and on MAE ($p_{\text{Holm}} < 10^{-10}$, δ between 0.35 and 0.48, medium to large); the Top-1 comparison is significant for all four models

Table S1.8: Paired architecture differences from the same cluster bootstrap. Each replicate scores both architectures on one resampled scenario set, so the interval carries the correlation between the two arms.

Comparison	Model	τ	MAE	Top-1 (pp)
$\mathcal{A}_E - \mathcal{A}_D$	DeepSeek	+0.163 [+0.070, +0.256]	-0.064 [-0.075, -0.052]	+16.9 [+10.1, +23.4]
	Gemini	+0.134 [+0.054, +0.216]	-0.061 [-0.071, -0.050]	+12.5 [+5.2, +20.0]
	GPT-OSS	+0.232 [+0.153, +0.310]	-0.072 [-0.084, -0.060]	+14.1 [+8.0, +19.8]
	Qwen	+0.198 [+0.106, +0.289]	-0.041 [-0.052, -0.031]	+16.6 [+10.2, +22.8]
$\mathcal{A}_H - \mathcal{A}_E$	DeepSeek	+0.591 [+0.511, +0.672]	-0.123 [-0.136, -0.110]	+37.2 [+31.3, +43.4]
	Gemini	+0.614 [+0.514, +0.714]	-0.111 [-0.125, -0.098]	+44.4 [+37.0, +51.7]
	GPT-OSS ^a	+0.628 [+0.547, +0.711]	-0.118 [-0.130, -0.106]	+44.5 [+38.5, +50.4]
	Qwen	+0.672 [+0.599, +0.742]	-0.122 [-0.138, -0.105]	+43.2 [+37.4, +48.8]
$\mathcal{A}_H - \mathcal{A}_D$	DeepSeek	+0.753 [+0.666, +0.842]	-0.187 [-0.197, -0.176]	+54.1 [+47.7, +60.4]
	Gemini	+0.747 [+0.654, +0.842]	-0.171 [-0.185, -0.158]	+56.9 [+50.3, +63.5]
	GPT-OSS ^a	+0.857 [+0.784, +0.928]	-0.189 [-0.202, -0.176]	+58.4 [+53.2, +63.6]
	Qwen	+0.870 [+0.797, +0.941]	-0.163 [-0.175, -0.151]	+59.7 [+54.2, +65.1]

Higher τ and Top-1 favor the first architecture; lower MAE favors it, so a negative MAE difference is an improvement. All 36 intervals exclude zero.

^a $n = 194$ scenarios; one scenario drops from the paired set on the $\mathcal{A}_H$ comparisons for GPT-OSS. Every other cell uses all 195.

but the effect size is small for GPT-OSS ($\delta = -0.175$) and Gemini ($\delta = -0.159$) and negligible for DeepSeek and Qwen ($\delta = -0.144$), since Top-1 is coarser than τ and both conditions already place the correct alternative first most of the time.

The extracted condition also differs from the true-parameter ceiling significantly on all three metrics for all four models, but on Kendall’s τ and Top-1 the effect size is negligible for every model ($|\delta| \leq 0.118$); only on MAE is the gap large ($\delta > 0.88$ for all four models). With $n = 195$ paired scenarios, a small and consistent shortfall is enough to reach significance without being practically large; the shortfall is a score-calibration gap rather than a ranking gap.

0.24 Alternative-Ordering Ablation: Full Statistics

Of the four models tested under the permutation protocol, two separated at the parameter level: Qwen and Gemini, with a separation of 0.054 and 0.047 respectively ($p \leq 0.0008$, Holm-adjusted $p \leq 0.006$ across the eight tests), both at the exact permutation floor of $1/1287$. No model separated on the top-1 alternative ($p \geq 0.24$). As a worked illustration of the mechanism, listing the alternatives in a different order moves Gemini’s estimate of a dwelling’s insulation and equipment age, quantities that have nothing to do with the order the setpoints are printed in; those shifted estimates are what the deterministic calculator absorbs before the ranking stage.

Table S1.13 gives the per-model McNemar statistics for $\mathcal{A}_D$ top-1 correctness under order reversal. Each reversed run is compared against all five shipped runs of the same model, so the discordant-pair counts are means over those five comparisons and the table reports the median and maximum p across them.

No model reaches significance on any comparison, with the smallest median p at 0.477. The discordant counts are large relative to the net difference in every model, which is what a run-to-run instability that swamps any ordering effect looks like: DeepSeek trades roughly 37 scenarios one way against 29 the other to net 8.2, rather than shifting a stable set of scenarios in one direction.

Kendall’s τ and Top-1 accuracy by prompt variant

Arch.	Model	Prompt variant			
		control	no_anchors	cot_scaffold	scale_0.10
<i>Kendall's τ</i>					
$\mathcal{A}_D$	DeepSeek	0.129	0.147	0.161	0.167
$\mathcal{A}_D$	Gemini	0.188	0.126	0.212	0.178
$\mathcal{A}_D$	GPT-OSS	0.053	0.100	0.041	−0.001
$\mathcal{A}_D$	Qwen	−0.052	0.024	0.092	0.015
$\mathcal{A}_E$	DeepSeek	0.304	—	0.180	0.246
$\mathcal{A}_E$	Gemini	0.325	—	0.245	0.250
$\mathcal{A}_E$	GPT-OSS	0.262	—	0.200	0.261
$\mathcal{A}_E$	Qwen	0.206	—	0.194	0.102
<i>Top-1 accuracy</i>					
$\mathcal{A}_D$	DeepSeek	0.394	0.458	0.363	0.412
$\mathcal{A}_D$	Gemini	0.352	0.361	0.368	0.350
$\mathcal{A}_D$	GPT-OSS	0.344	0.368	0.328	0.320
$\mathcal{A}_D$	Qwen	0.289	0.371	0.333	0.329
$\mathcal{A}_E$	DeepSeek	0.552	—	0.367	0.478
$\mathcal{A}_E$	Gemini	0.477	—	0.410	0.416
$\mathcal{A}_E$	GPT-OSS	0.454	—	0.440	0.489
$\mathcal{A}_E$	Qwen	0.474	—	0.399	0.407

Mean over five runs on 195 Test scenarios per cell, except $\mathcal{A}_D$ control and no_anchors, which use ten runs: no_anchors was extended from five to ten runs to give a borderline control-vs-no_anchors comparison enough power to resolve, and control was matched to ten for a balanced comparison. Gemini cells use three runs (rationale below). Bold marks the best variant within each architecture–model row group. $\mathcal{A}_E$ has no no_anchors variant because its shipped prompt contains no anchors.

Table S1.9: Prompt-sensitivity ablation. Within each model, every $\mathcal{A}_E$ configuration remains above every $\mathcal{A}_D$ configuration on Kendall’s τ , so no prompt perturbation reverses the architecture ordering for a fixed model.

Table S1.10: Prompt ablation: Friedman omnibus tests for $\mathcal{A}_E$, corrected across the full 16-test family (both architectures, all four models, both metrics).

Model	Metric	χ^2_2	raw p	p_{Holm}
DeepSeek	Kendall’s τ	1.51	0.470	0.939
Gemini	Kendall’s τ	3.31	0.192	0.766
GPT-OSS	Kendall’s τ	3.38	0.184	1.000
Qwen	Kendall’s τ	5.71	0.057	0.575
DeepSeek	Top-1	35.71	1.8×10^{-8}	2.8×10^{-7}
Gemini	Top-1	9.08	0.011	0.117
Qwen	Top-1	9.87	0.007	0.086

Holm adjustment is step-down with monotonicity enforced, so a test with a larger raw p cannot receive a smaller adjusted p . The GPT-OSS Top-1 omnibus is part of the 16-test family but is not reported here.

Table S1.11: Prompt ablation: significant pairwise comparisons (Holm-corrected post-hoc, run only within omnibus families that survived correction).

Arch.	Model	Comparison	Value	p_{Holm}	Cliff’s δ
$\mathcal{A}_E$	DeepSeek	control vs. cot_scaffold (Top-1)	0.552 $\rightarrow$ 0.367	2.9×10^{-7}	0.276 (small)
$\mathcal{A}_E$	DeepSeek	control vs. scale_0.10 (Top-1)	0.552 $\rightarrow$ 0.478	0.003	0.116 (negligible)
$\mathcal{A}_E$	DeepSeek	cot_scaffold vs. scale_0.10 (Top-1)	0.367 $\rightarrow$ 0.478	2.2×10^{-4}	−0.190 (small)
$\mathcal{A}_D$	Qwen	control vs. cot_scaffold (τ)	−0.052 $\rightarrow$ 0.092	3.8×10^{-4}	−0.165 (small)
$\mathcal{A}_D$	Qwen	control vs. scale_0.10 (τ)	−0.052 $\rightarrow$ 0.015	0.028	−0.100 (negligible)

All five comparisons are significant under the two-stage Holm correction. Cliff’s δ sign follows $P(\text{config}_i > \text{config}_j) - P(\text{config}_i < \text{config}_j)$; negative values indicate the second-listed configuration scores higher.

0.25 Imputed Robustness Check

The main results use per-run-then-average aggregation: failed scenarios are excluded from their run’s metric computation, and metrics are averaged across five runs. To test sensitivity to this exclusion, Table S1.14 reports an imputed alternative at per-run granularity: within each run, scenarios that carry a failed score receive the scale midpoint (0.5) on the affected criterion scores before that run’s metrics are computed, and every run evaluates the full 195-scenario test set.

0.26 Overall Results: Numerical Values

Table S1.15 reports the full numerical results for each metric, architecture, and model, with five-run means and standard deviations.

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Parameter-source ablation

Model	Parameter source	τ	Top-1	MAE
GPT-OSS	True (ceiling)	1.000	1.000	0.000
GPT-OSS	LLM-extracted	0.921	0.943	0.046
GPT-OSS	Corpus median	0.641	0.769	0.122
Qwen	True (ceiling)	1.000	1.000	0.000
Qwen	LLM-extracted	0.897	0.913	0.066
Qwen	Corpus median	0.641	0.769	0.122
DeepSeek	True (ceiling)	1.000	1.000	0.000
DeepSeek	LLM-extracted	0.904	0.913	0.041
DeepSeek	Corpus median	0.641	0.769	0.122
Gemini	True (ceiling)	1.000	1.000	0.000
Gemini	LLM-extracted	0.921	0.928	0.047
Gemini	Corpus median	0.641	0.769	0.122

All conditions scored by the same reference calculators over the same 195 test scenarios; the conditions differ only in the source of the withheld engineering parameters. The true-parameter condition reproduces the reference ranking exactly by construction, which serves as a correctness check on the harness. The corpus-median condition uses one constant per parameter and no model output, so it is identical across models, and the four identical 0.641 / 0.769 / 0.122 rows serve as a second correctness check on the harness. Scenarios whose extraction failed are excluded from that condition rather than defaulted; the extracted condition scored 194 of 195 scenarios for GPT-OSS and 195 for the other three models.

Table S1.12: Ranking accuracy of $\mathcal{A}_H$ with true, LLM-extracted, and corpus-median hidden parameters. Significance testing on these conditions follows.

Table S1.13: Alternative-ordering McNemar tests for $\mathcal{A}_D$ top-1 correctness, per model. Discordant counts are means across the five shipped-versus-reversed comparisons. Positive net values favour the reversed condition. No value is pooled across models.

Model	n	Discordant pairs		Net	McNemar p		Top-1 agreement
		Ship ✓	Rev ✓		Median	Max	
DeepSeek V4 Flash	193	37.2	29.0	−8.2	0.477	0.807	0.482
Gemini 3.5 Flash	195	16.4	17.8	+1.4	0.856	1.000	0.748
GPT-OSS 20B	193	30.2	33.6	+3.4	0.615	0.902	0.523
Qwen3.5 9B	190	32.6	34.8	+2.2	0.712	1.000	0.479

Ship ✓: scenarios the shipped order ranked correctly and the reversed order did not. Rev ✓: the converse. Top-1 agreement is the share of scenarios on which the two orders selected the same alternative, correct or not.

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Table S1.14: Comparison of per-run-then-average (Method A) vs per-run imputed (0.5 midpoint) aggregation. $\mathcal{A}_D$ and $\mathcal{A}_E$ values are means across all four models; $\mathcal{A}_H$ is broken out per model, because imputation moves one model there and leaves the other three unchanged.

Architecture	Metric	Method A	Imputed (per-run)
$\mathcal{A}_D$ (4-model mean)	τ	0.093	0.094
$\mathcal{A}_D$ (4-model mean)	MAE	0.232	0.232
$\mathcal{A}_D$ (4-model mean)	Top-1 (%)	34.1	34.2
$\mathcal{A}_E$ (4-model mean)	τ	0.274	0.274
$\mathcal{A}_E$ (4-model mean)	MAE	0.172	0.173
$\mathcal{A}_E$ (4-model mean)	Top-1 (%)	49.0	49.1
$\mathcal{A}_H$ (DeepSeek)	τ	0.897	0.897
$\mathcal{A}_H$ (DeepSeek)	MAE	0.045	0.045
$\mathcal{A}_H$ (DeepSeek)	Top-1 (%)	90.8	90.7
$\mathcal{A}_H$ (Gemini)	τ	0.923	0.923
$\mathcal{A}_H$ (Gemini)	MAE	0.048	0.048
$\mathcal{A}_H$ (Gemini)	Top-1 (%)	93.1	93.1
$\mathcal{A}_H$ (GPT-OSS)	τ	0.897	0.787
$\mathcal{A}_H$ (GPT-OSS)	MAE	0.052	0.077
$\mathcal{A}_H$ (GPT-OSS)	Top-1 (%)	91.7	83.7
$\mathcal{A}_H$ (Qwen)	τ	0.880	0.877
$\mathcal{A}_H$ (Qwen)	MAE	0.072	0.072
$\mathcal{A}_H$ (Qwen)	Top-1 (%)	89.7	89.5

Mean scenarios imputed per run: DeepSeek $\mathcal{A}_D$ 1.4, DeepSeek $\mathcal{A}_H$ 0.2, Gemini $\mathcal{A}_E$ 2.0, GPT-OSS $\mathcal{A}_H$ 23.4, Qwen $\mathcal{A}_D$ 0.4, Qwen $\mathcal{A}_H$ 0.4; all other cells impute none. GPT-OSS imputes 20–26 scenarios per run on $\mathcal{A}_H$, 12% of its test set.

Imputed scenarios receive 0.5 on all four criterion scores; their ranks follow the pipeline’s deterministic tie-break (appearance order).

Bold marks the only cells that move materially. $\mathcal{A}_D$ and $\mathcal{A}_E$ are reported as four-model means because no model’s cell moves by more than 0.006 in either architecture.

Metric	Gemini			DeepSeek		
	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$
Kendall's τ	0.176 ± 0.014	0.310 ± 0.019	0.923 ± 0.002	0.144 ± 0.041	0.307 ± 0.036	0.897 ± 0.006
Top-1 (%)	36.2 ± 1.5	48.7 ± 2.0	93.1 ± 0.3	36.7 ± 3.1	53.5 ± 3.3	90.8 ± 1.0
MAE	0.219 ± 0.002	0.158 ± 0.001	0.048 ± 0.001	0.231 ± 0.003	0.168 ± 0.001	0.045 ± 0.004
RMSE	0.295 ± 0.002	0.231 ± 0.002	0.101 ± 0.001	0.302 ± 0.003	0.237 ± 0.001	0.092 ± 0.011

Metric	GPT-OSS			Qwen		
	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$	$\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$
Kendall's τ	0.041 ± 0.045	0.272 ± 0.040	0.897 ± 0.007	0.010 ± 0.039	0.208 ± 0.037	0.880 ± 0.013
Top-1 (%)	33.3 ± 2.4	47.4 ± 2.1	91.7 ± 0.7	30.0 ± 3.0	46.6 ± 1.6	89.7 ± 1.2
MAE	0.241 ± 0.002	0.169 ± 0.002	0.052 ± 0.001	0.235 ± 0.002	0.194 ± 0.003	0.072 ± 0.002
RMSE	0.306 ± 0.002	0.242 ± 0.003	0.101 ± 0.004	0.295 ± 0.002	0.261 ± 0.004	0.162 ± 0.005

Table S1.15: Overall performance by model (5-run mean $\pm$ std, conditional on successful extraction over the 195 test scenarios). Bolded values indicate the best architecture per metric per model.

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